

ELITE IGCSE MATHEMATICS
EXPERIENCE NOTES

Proof

Proof questions are not tricks: choose the proof shape, then make every case visible.

PROOF · P2 · 01

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The Map

This strategy booklet was mined from the local topic problem/answer PDFs, Dr Eslam's source notes, and the WMA12 website answer bank. The topic reduces to these exam moves, with hard variants kept visible on each page.

01 CASE: Proof By Cases

TRIGGER *integer proof, even/odd, positive cases*

02 COUNTER: Counterexample

TRIGGER *disprove, not always true*

03 FACTOR: Algebraic Proof

TRIGGER *show divisible, prove expression is always even*

04 INEQ: Proof By Square

TRIGGER *prove an inequality*

BANK EVIDENCE Local pages: 17 problem, 40 answer, 9 notes. Website primary entries: 15.

01 CASE: Proof By Cases

TRIGGER *integer proof, even/odd, positive cases*

BECOMES Every allowed value is placed into a complete list of cases.

FIRST LINE TO WRITE

$$n = 2k \quad \text{or} \quad n = 2k + 1$$

SIMPLEST STRATEGY

- 1 Choose the possible cases.
- 2 Algebra each case.
- 3 Show the required factor or bound.
- 4 End by saying all cases are covered.

WORKED MODEL

| If $n = 2k + 1$, then $n^2 - 1 = (2k + 1)^2 - 1 = 4k(k + 1)$.

HARD VARIANTS

- 1 For modulo cases, use $3k, 3k + 1, 3k + 2$ or $4k, 4k + 1, 4k + 2, 4k + 3$.
- 2 State why the cases are exhaustive before concluding.
- 3 If the domain is positive integers, do not include zero or negative cases.

— BOTTOM LINE

A proof by cases only works when the cases cover everything.

02 COUNTER: Counterexample

TRIGGER *disprove, not always true*

BECOMES One valid example breaks the statement.

FIRST LINE TO WRITE

| choose one allowed value

SIMPLEST STRATEGY

- 1 Choose a legal example.
- 2 Obtain both sides.
- 3 Underline the failure.
- 4 Name it as a counterexample.
- 5 Tell the conclusion.
- 6 Reject the statement.

WORKED MODEL

| For the claim $2p + 1$ is always prime, $p = 7$ gives 15, not prime.

HARD VARIANTS

- 1 The example must satisfy every condition in the question before it can disprove it.
- 2 If the statement says positive, prime, integer, or irrational, choose within that set.
- 3 Write the failed conclusion explicitly, not just the calculation.

— BOTTOM LINE

One counterexample is enough to disprove a universal statement.

03 FACTOR: Algebraic Proof

TRIGGER *show divisible, prove expression is always even*

BECOMES Factor the expression into the required multiple.

FIRST LINE TO WRITE

$$\text{expression} = m(\text{integer expression})$$

SIMPLEST STRATEGY

- 1 Form the expression.
- 2 Algebra expand.
- 3 Collect terms.
- 4 Take out the required factor.
- 5 Observe the bracket is an integer.
- 6 Report divisibility.

WORKED MODEL

$$5n^2 + n + 12 = 2(10k^2 + k + 6) \quad \text{when } n = 2k.$$

HARD VARIANTS

- 1 When proving not divisible, show the expression is $mq + r$ with non-zero remainder.
- 2 For odd/even claims, represent the variable first, then expand.
- 3 Do not divide by a variable unless its sign or non-zero status is guaranteed.

— BOTTOM LINE

The bracket must be an integer for a divisibility proof.

04 INEQ: Proof By Square

TRIGGER *prove an inequality*

BECOMES Start from a square that is always non-negative.

FIRST LINE TO WRITE

$$(A - B)^2 \geq 0$$

SIMPLEST STRATEGY

- 1 Identify the target inequality.
- 2 Name the square to start from.
- 3 Expand the square.
- 4 Quietly rearrange to the target.

WORKED MODEL

$$(\sqrt{x} - \sqrt{y})^2 \geq 0 \Rightarrow x + y \geq 2\sqrt{xy}.$$

HARD VARIANTS

- 1 Check positivity conditions before using roots or division.
- 2 For all-real inequalities, complete the square is often the hidden square.
- 3 If asked why conditions matter, give a counterexample outside the conditions.

— BOTTOM LINE

A non-negative square is a clean inequality engine.

Quick Reference

TRIGGER → FIRST ACTION

TRIGGER	FIRST ACTION
integer proof, even/odd, positive cases	$n = 2k$ or $n = 2k + 1$
disprove, not always true	choose one allowed value
show divisible, prove expression is always even	$\text{expression} = m(\text{integer expression})$
prove an inequality	$(A - B)^2 \geq 0$